

Git & Amazon Web Services

Git

History

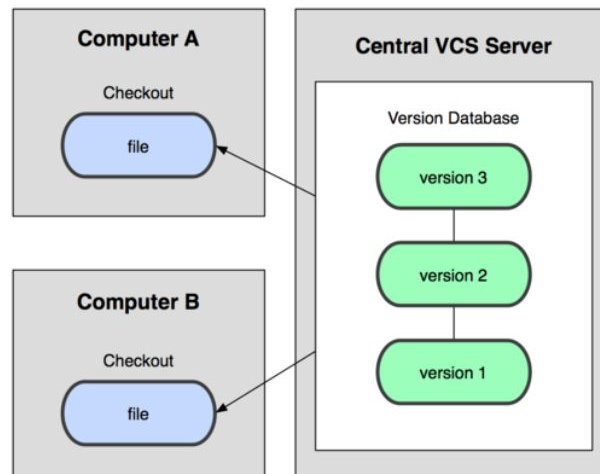
- Linux needed some sort of Version Control System so Linus Torvalds (who created Linux) developed it in 2005
 - Previously used something called BitKeeper
- Linus developed it for speed, efficiency, and non-linear development (parallel branches)

What is Git?

- It is a Version Control System (VCS)
 - System that keeps track of changes made to files
- Very important when working in larger software engineering teams, deploying versions of software, and ensuring changes do not get deleted!

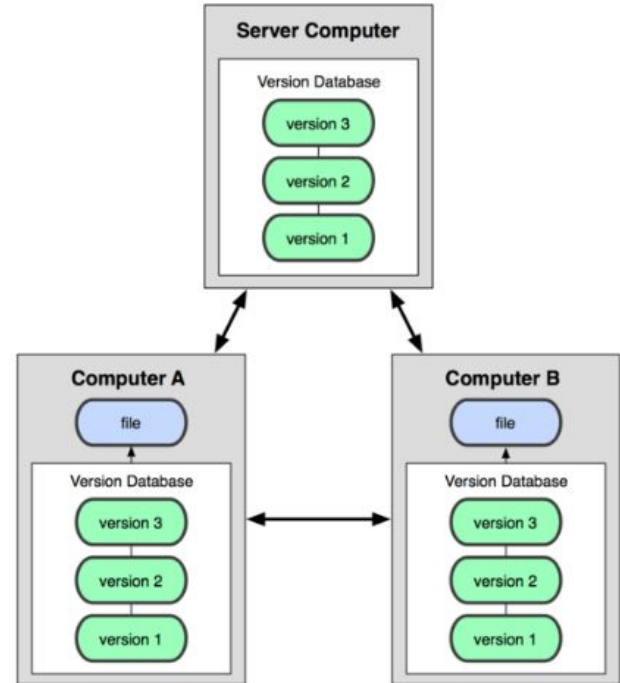
Centralized VCS

- Some examples are CVS and Subversion (SVN)
 - Google code was a SVN and a lot of government employees and contractors are too scared to switch from SVN because it is “stable” and their code base is already there
- Repository is on a central server and client only has one version of branch



Distributed VCS

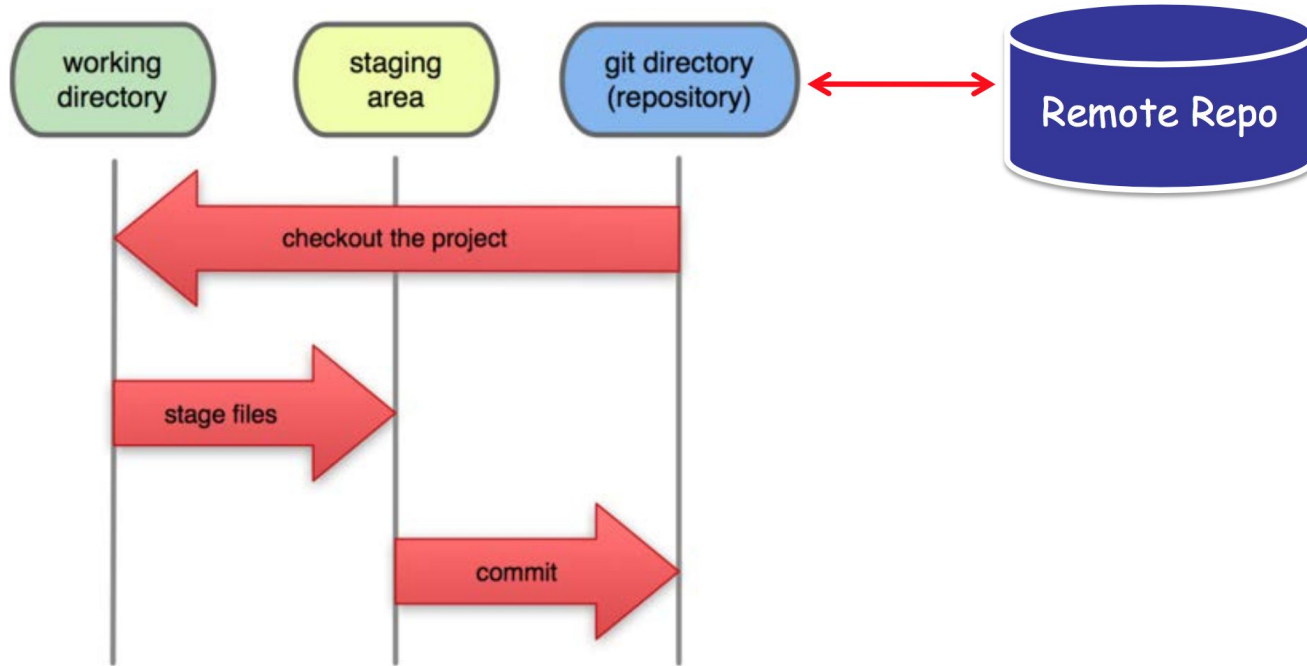
- Git
 - The full repository is local and has full history
 - Server is involved but you can commit offline
 - Can push and pull between repos



Git Basics

- Nearly every operation is local - which makes it fast!
- Data managed with integrity
 - 40 character SHA-1 hash (also used in cryptography) computed and assigned to every commit, branch and tag
 - Makes Git aware of hacking attempts or repo corruption
- Git only adds data for the most part, nothing is deleted

Git Basics (cont.)



Git Setup

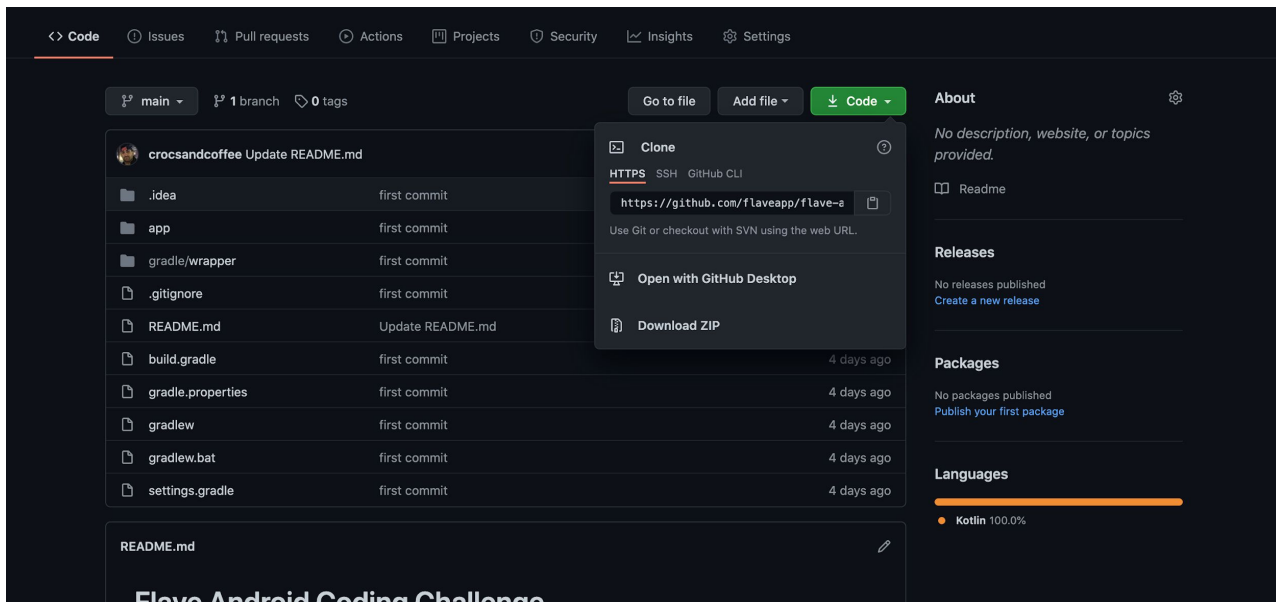
- Setup properties globally
 - `$git config --global user.name "Jal Irani"`
 - `$git config --global user.email jirani@towson.edu`
 - `$git config user.name` - will give Jal Irani

Create a New Git Repository

- Create a new repo
 - `$cd projects`
 - `$git init` - may want to add a `.gitignore` file
 - `$git add .` (. Adds the entire current directory with subdirectories)
 - `$git commit -m "Initial commit"`
 - `$git push origin <branch-name>`

Cloning an existing Repo

- `$git clone http://www.someurl.git`



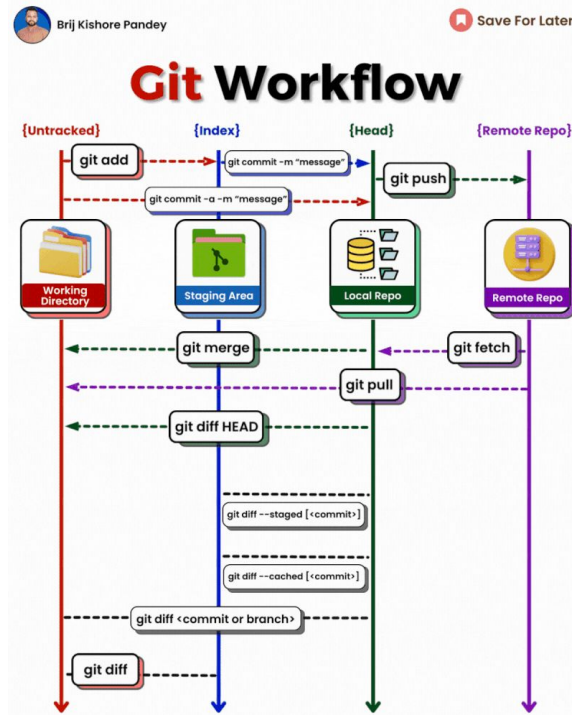
Important Git Commands

- `$git status`
 - Gives the current state of the repository
 - Files that are added, removed, or edited
- `$git add <file/dir>`
 - Adds untracked files to be tracked
 - Add tracked file to staging area
 - Mods made to file after “git add” need to be re added before commit

Important Git Commands (cont.)

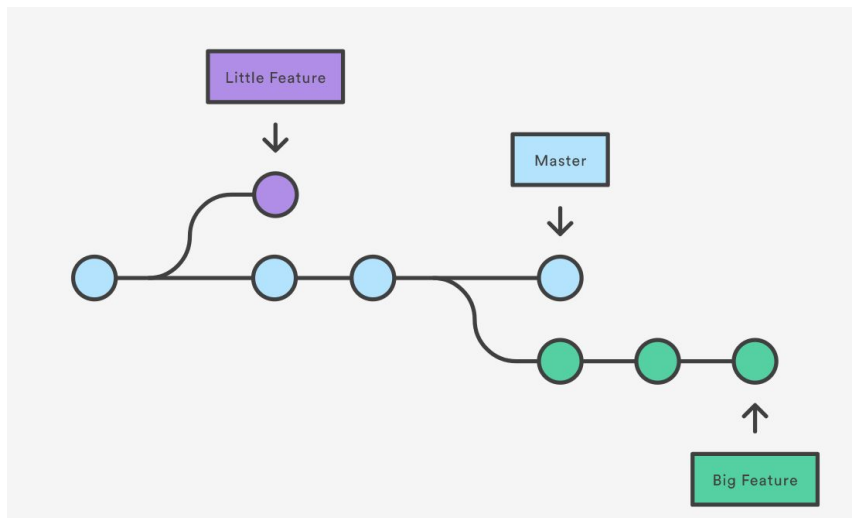
- `$git commit -m "some message"`
 - This will commit the staged files to your local repository
- `$git push origin <branch>`
 - This will push to a remote branch so that your repository exists somewhere besides your local machine
 - GitHub
 - Bitbucket
 - Etc.
- `$git revert HEAD`
 - Reverts the most recent commit

Git Flow with Commands



What is a Branch in Git?

- Branches serve two purposes
 - Allow team members to work concurrently on the same project
 - The easiest way of adding new features to a project without disrupting workflow



Branching Explained

- Every Git repository will have a Master/Main branch
 - Master should always include the current working version
 - “Protected” or “release” branch
- Most repositories will also have “feature” branches
 - These branches will “branch” from Master and will incorporate new features into the the product
 - When the feature is completed (tested and working correctly), this branch will be merged into Master

Branching Example

- We have a Node.js server with an endpoint called “/example”
- We want to add a new endpoint called “/branch”
- If this product is in production, best practice would be
 - Branch from Master and create a branch called “jji-branch-feature”
 - Add the endpoint and test thoroughly
 - Merge my feature branch back into Master
 - Redeploy Master so it is a seamless update

View Branches in Git

- To view all branches that are checked out locally and exist remotely use the command:
 - `$git branch -a -v`

```
Jals-MacBook-Pro:sleepin jalirani$ git branch -a -v
* master          42c3178 background modes fixed, schools fixed, and no termination notification
remotes/origin/HEAD -> origin/master
remotes/origin/master 42c3178 background modes fixed, schools fixed, and no termination notification
Jals-MacBook-Pro:sleepin jalirani$
```

Create a Branch in Git

- Use “git checkout -b <branch-name>” to create a new branch

```
Jals-MacBook-Pro:git_node_example jalirani$ git checkout -b "jji-feature1"  
Switched to a new branch 'jji-feature1'
```

- Use “git checkout <branch-name>” to check out a different branch
- To remove a branch use “git branch -d <branch-name>”

One Thing to Note

- Let's say you are working on your own branch when another branch gets pulled into Master?
 - That means you are 1 commit behind HEAD (aka Master is getting updated - so your feature branch is currently out of date)
- You need to pull from master!

```
Jals-MacBook-Pro:sleepin jalirani$ git pull origin master
From https://bitbucket.org/JalIrani/sleepin
 * branch                master      -> FETCH_HEAD
Already up-to-date.
```

Git References

- <https://services.github.com/>
- <https://git-scm.com/>

Amazon Web Services

What Exactly is AWS?

- AWS provides on-demand cloud computing platforms with either free or paid tier subscriptions
- Most options will provide you an operating system, CPUs & GPUs, RAM, hard-disk space, etc.
- AWS is hosted on different server farms (distributed systems) around the world, so there will not be reliability issues in terms of “servers going down”

Alternate Options

- The two main competitors are
 - Microsoft Azure
 - Google Cloud
- The main reason developers opt for Azure or Google Cloud is because of pricing
- AWS is the main leader in the space and widely used in industry today, but AWS costs can rack up a hefty fee if traffic is high on your website

AWS Breakdown

There are two main “instances” that we will focus on for this class

- Amazon EC2
- Amazon S3

Amazon EC2

- This stands for Amazon Elastic Compute Cloud
 - Amazon defines this as a web service that provides resizable compute capacity in the cloud
- What does that mean?
 - This will typically be where Node.js server, database, etc. will live
 - The “elastic” component will allow developers to scale
 - When there is a dip in requests, developers can use a smaller amount of computing resources or increase accordingly
 - Seen as more flexible, but slower than S3

Amazon S3

- Amazon defines S3 as intentionally built with a minimal feature set providing the ability to read, write, and delete object
- What does that mean?
 - Normally you will store static content on S3
 - Since web servers are typically slow and file servers are fast
 - If you have a single monolithic web application served from an EC2 instance, for example, this means that to view a page, you need to run some code to generate that HTML.
 - If you deploy your front-end code to S3, however, that file can be directly downloaded by the web browser MUCH faster.

AWS Intro

- https://aws.amazon.com/training/intro_series/

Separate Deployment

- With a large scale application, you typically deploy front-end to S3 and back-end to EC2 for speed and efficiency
- No issue for connectivity because requests (GET and POST) will be routed to the back-end API and the backend will give a response to the front-end
- Only difference is they will not be in the same directory like you typically see on your local machine

Order of Operations

- Using the AWS console, launch your instance
 - Then FTP/SCP your server code to the AWS IP address (OR)
 - If you are smart, you will store all of your server code in Git so all you will have to do is ssh into your instance and git clone URL
 - Setup some solid CI/CD :)
- Start the server
- Donezo

Hosting Tutorial

<https://www.youtube.com/watch?v=E78VKAPwmDM>

Things to Note

- These tutorials fail to mention a lot of things with server-side programming
- One big thing that is left out is that a server must be started on AWS and then most likely the developer will want to exit out of the terminal
 - This will cause the server to stop running
- “Nohup” command will help with this:
<http://linux.101hacks.com/unix/nohup-command/>
 - I.e. “nohup node server.js” to ensure your server runs without hangup

If You are into DevOps and Cloud Computing

- Definitely not my forte, but check out:
 - <https://www.coursera.org/specializations/cloud-computing>
 - UIUC is one of the best schools in the world for distributed systems and cloud computing world-wide
 - There is more to cloud-computing then just AWS so very cool to check out